The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Digital Image Processing

Lab N0#01

Title: Python Setup, Fundamentals, and Imports

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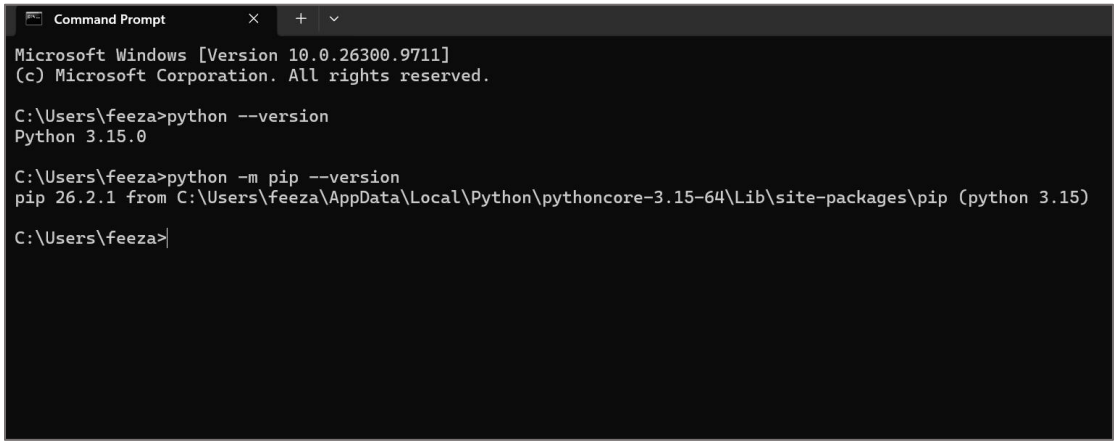
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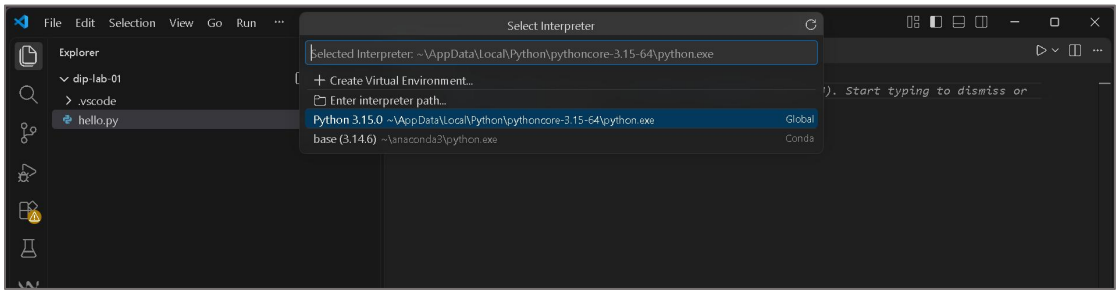
■ python and pip Verification

A Windows Command Prompt terminal window showing the execution of python --version and python -m pip --version. It confirms that Python 3.15.0 and pip 26.2.1 are installed on the local system under the user path C:\Users\feeza.



■ VS Code Interpreter Selection

The Visual Studio Code editor interface with the "Select Interpreter" quick-pick palette open. It shows the active selection configured to the global Python 3.15.0 environment (pythoncore-3.15-64\python.exe) over the base Conda environment for the dip-lab-01 project workspace.



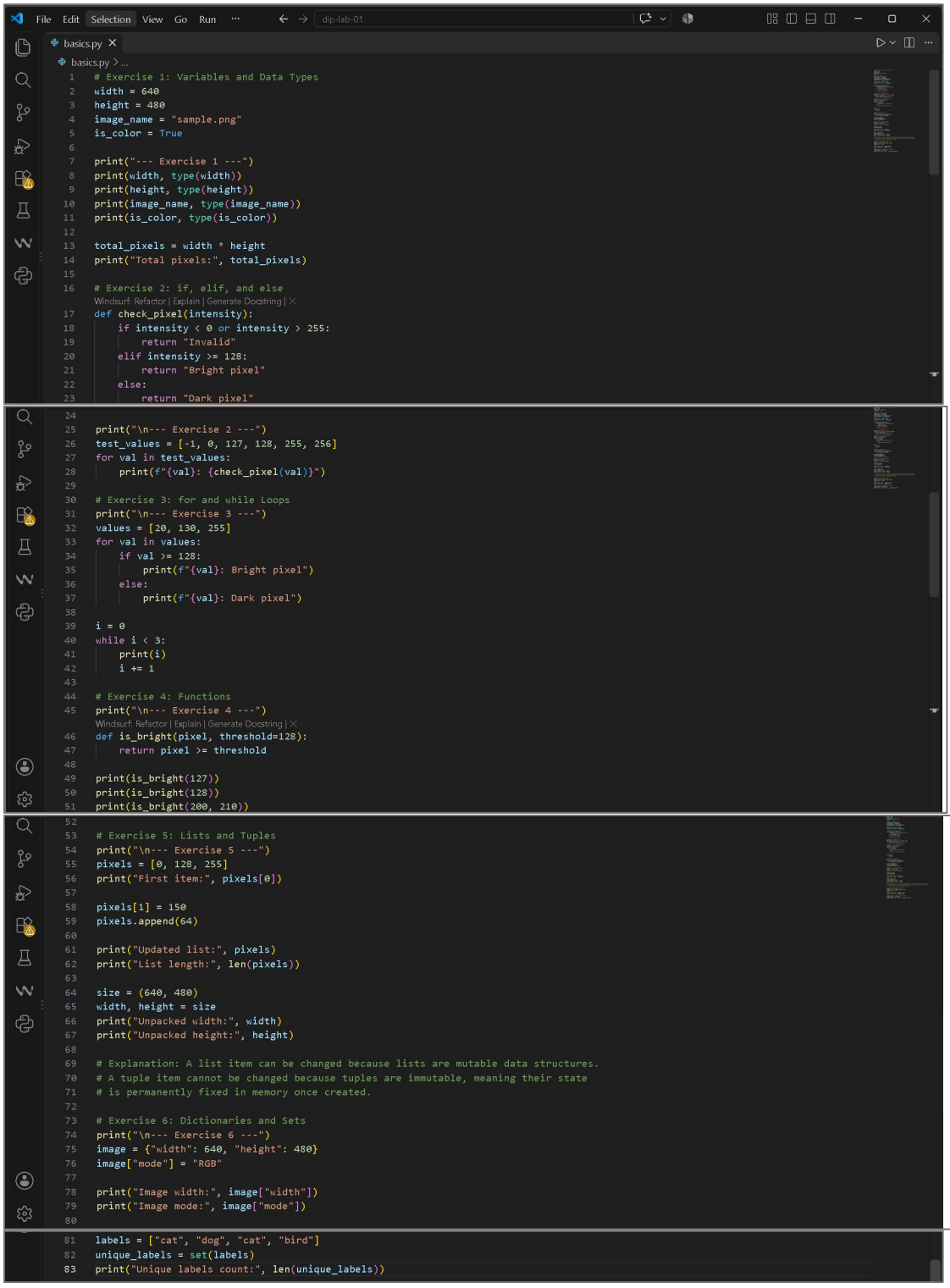
■ hello.py execution

The VS Code editor displaying hello.py containing a single line: print('Hello, Digital Image Processing!'). The integrated PowerShell terminal beneath shows the successful execution command python hello.py and its corresponding console output.



■ basics.py code and Terminal Output

The basics.py script covers fundamental Python constructs for image processing, including variable typing, pixel resolution calculations, conditional intensity thresholding, and loop iterations. It tests list mutability, tuple unpacking, dictionary metadata, and set deduplication, with the terminal output confirming expected values across all six exercises.



```python
# Exercise 1: Variables and Data Types
width = 640
height = 480
image_name = "sample.png"
is_color = True

print("--- Exercise 1 ---")
print(width, type(width))
print(height, type(height))
print(image_name, type(image_name))
print(is_color, type(is_color))

total_pixels = width * height
print("Total pixels:", total_pixels)

# Exercise 2: if, elif, and else
def check_pixel(intensity):
    if intensity < 0 or intensity > 255:
        return "Invalid"
    elif intensity >= 128:
        return "Bright pixel"
    else:
        return "Dark pixel"


print("\n--- Exercise 2 ---")
test_values = [-1, 0, 127, 128, 255, 256]
for val in test_values:
    print(f"{val}: {check_pixel(val)}")

# Exercise 3: for and while Loops
print("\n--- Exercise 3 ---")
values = [20, 130, 255]
for val in values:
    if val >= 128:
        print(f"{val}: Bright pixel")
    else:
        print(f"{val}: Dark pixel")

i = 0
while i < 3:
    print(i)
    i += 1

# Exercise 4: Functions
print("\n--- Exercise 4 ---")
def is_bright(pixel, threshold=128):
    return pixel >= threshold

print(is_bright(127))
print(is_bright(128))
print(is_bright(200, 210))


# Exercise 5: Lists and Tuples
print("\n--- Exercise 5 ---")
pixels = [0, 128, 255]
print("First item:", pixels[0])

pixels[1] = 150
pixels.append(64)

print("Updated list:", pixels)
print("List length:", len(pixels))

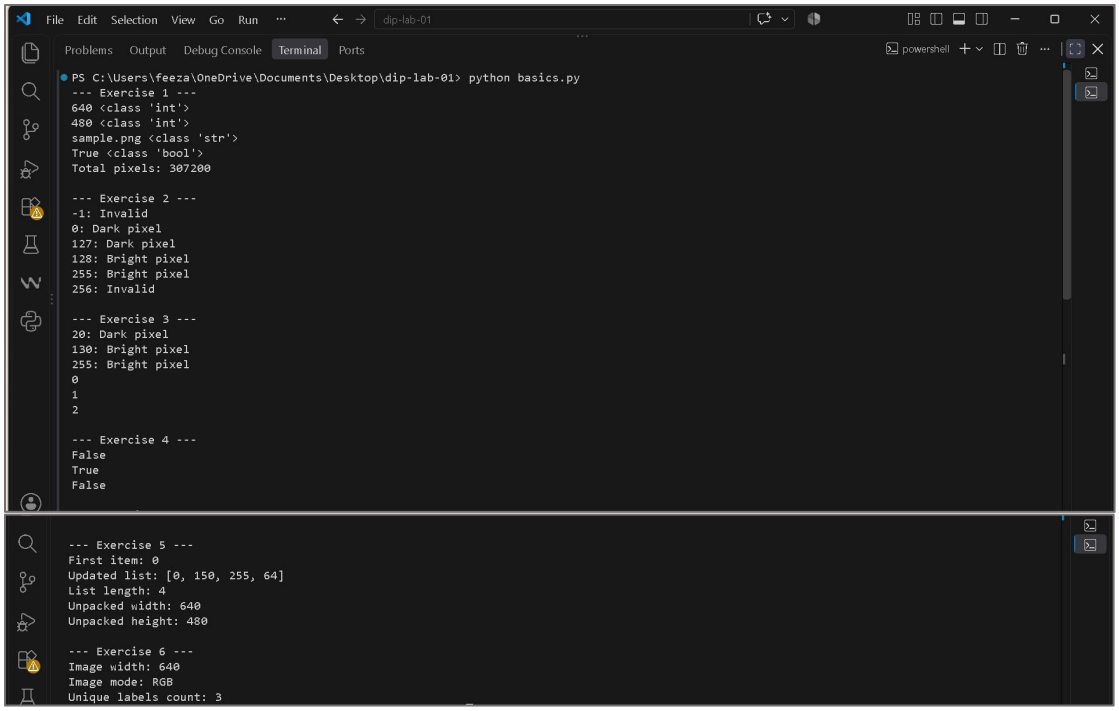
size = (640, 480)
width, height = size
print("Unpacked width:", width)
print("Unpacked height:", height)

# Explanation: A list item can be changed because lists are mutable data structures.
# A tuple item cannot be changed because tuples are immutable, meaning their state
# is permanently fixed in memory once created.

# Exercise 6: Dictionaries and Sets
print("\n--- Exercise 6 ---")
image = {"width": 640, "height": 480}
image["mode"] = "RGB"

print("Image width:", image["width"])
print("Image mode:", image["mode"])

labels = ["cat", "dog", "cat", "bird"]
unique_labels = set(labels)
print("Unique labels count:", len(unique_labels))
```



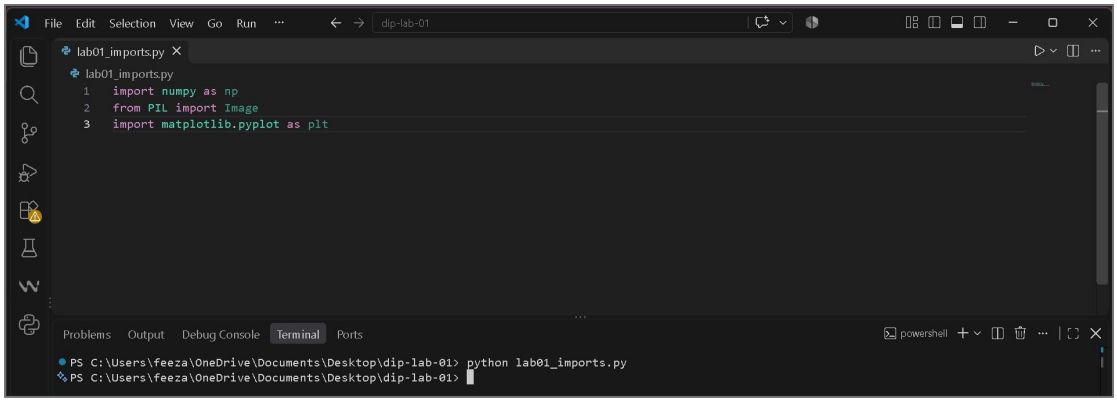
■ Package Installation Verification

PowerShell terminal showing the command python -m pip show numpy pillow matplotlib typed at the prompt in the dip-lab-01 directory to query installed digital image processing libraries.Terminal output displaying package metadata details (installation path under site-packages and dependency requirements such as NumPy, Pillow, fonttools, and cycler) confirming that the image processing toolchain is properly installed



■ lab01_imports.py Execution

VS Code editor with lab01_imports.py open showing the three core library imports (import numpy as np, from PIL import Image, and import matplotlib.pyplot as plt). The terminal below shows python lab01_imports.py executing cleanly with no errors, confirming all required modules load properly.



■ Python vs. pip and Import

Python is the core programming language and runtime environment that executes code, whereas pip is its standard package manager used to download and install external libraries from the Python Package Index. The import statements are utilized to bring those external libraries (like NumPy and Pillow) into the current script so their specialized image-processing functions and arrays can be used.